

DP-ES: Differentially Private Evolution Strategies for Prompt Optimization

Ziniu Liu¹, Aiping Li^{1,*}, Yue Han¹, Han Yu¹, Junjian Zhang¹, Dong Zhu¹,
Changjian Li¹, Shiqiang Zhang²

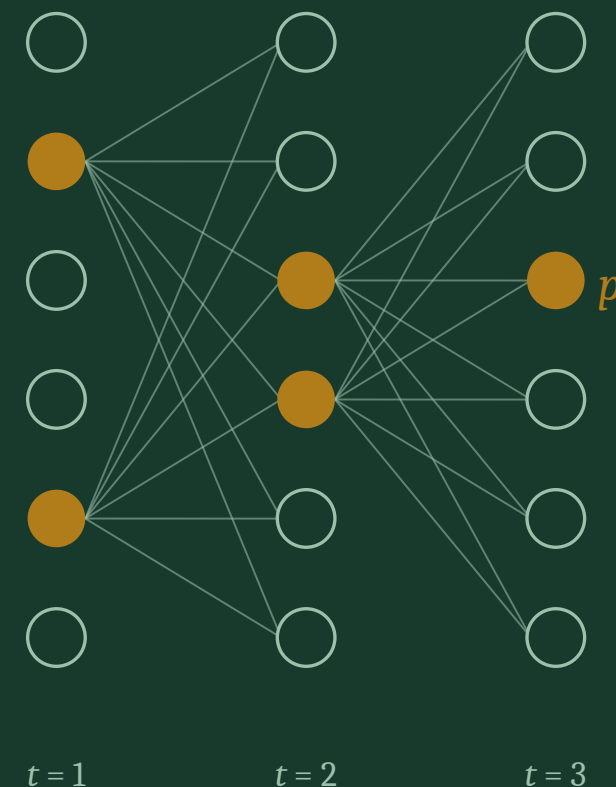
¹ National University of Defense Technology

² CRRC Zhuzhou Electric Locomotive Research Institute Co., Ltd.,
China Academy of Railway Sciences

* Corresponding author: liaiping@nudt.edu.cn



github.com/StephCpa/dp-es



$K = 6$ candidates • $M = 2$ parents • $T = 3$ rounds

Outline

- 1 Motivation**
Why prompt optimization on sensitive data needs formal privacy
- 2 Diagnosis**
Why token-level DP prompt construction is unstable under tight budgets
- 3 Method**
DP-ES: privacy-free mutation, private scoring, post-processed selection
- 4 Results**
Accuracy, stability, efficiency, ablations, memorization stress test
- 5 Scope**
What the evidence does and does not establish

THE TALK IN ONE SENTENCE

Evolving a population of complete prompts — instead of building one prompt token by token — makes differentially private prompt optimization markedly more robust at $\epsilon \leq 1$.

88.1%

GSM8K

~9×

lower std

2.5×

faster

Prompt optimizers can memorize private data

WHAT A NAIVE OPTIMIZER CAN OUTPUT

“Remember how patient #12847 had atrial fibrillation ...”

An illustrative optimized prompt that copies a record from the optimization set.



Membership-inference and extraction attacks on LLMs make this a practical risk (Wang et al., 2023; Fu et al., 2023; Shanmugarasa et al., 2025).



Sensitive optimization data

Optimizers such as APE, OPRO and TextGrad tune prompts on task data — e.g., patient records or financial logs — with no privacy guarantee.



Black-box LLM APIs

Frontier models are API-only, so weight-based private training (DP-SGD, DP-LoRA) does not apply.



What we need

Formal record-level (ϵ , δ)-DP for the released prompt — and an optimizer that stays robust under DP noise.

Goal: optimize a discrete prompt under DP

OBJECTIVE

Private dataset $D = \{(x_i, y_i)\}_{i=1}^n$, prompt space P :

$$p^* = \arg \max_{p \in P} f(p, D)$$

PRIVACY REQUIREMENT

The optimizer M is (ϵ, δ) -DP if, for all neighbouring datasets $D \sim D'$ and all sets S ,

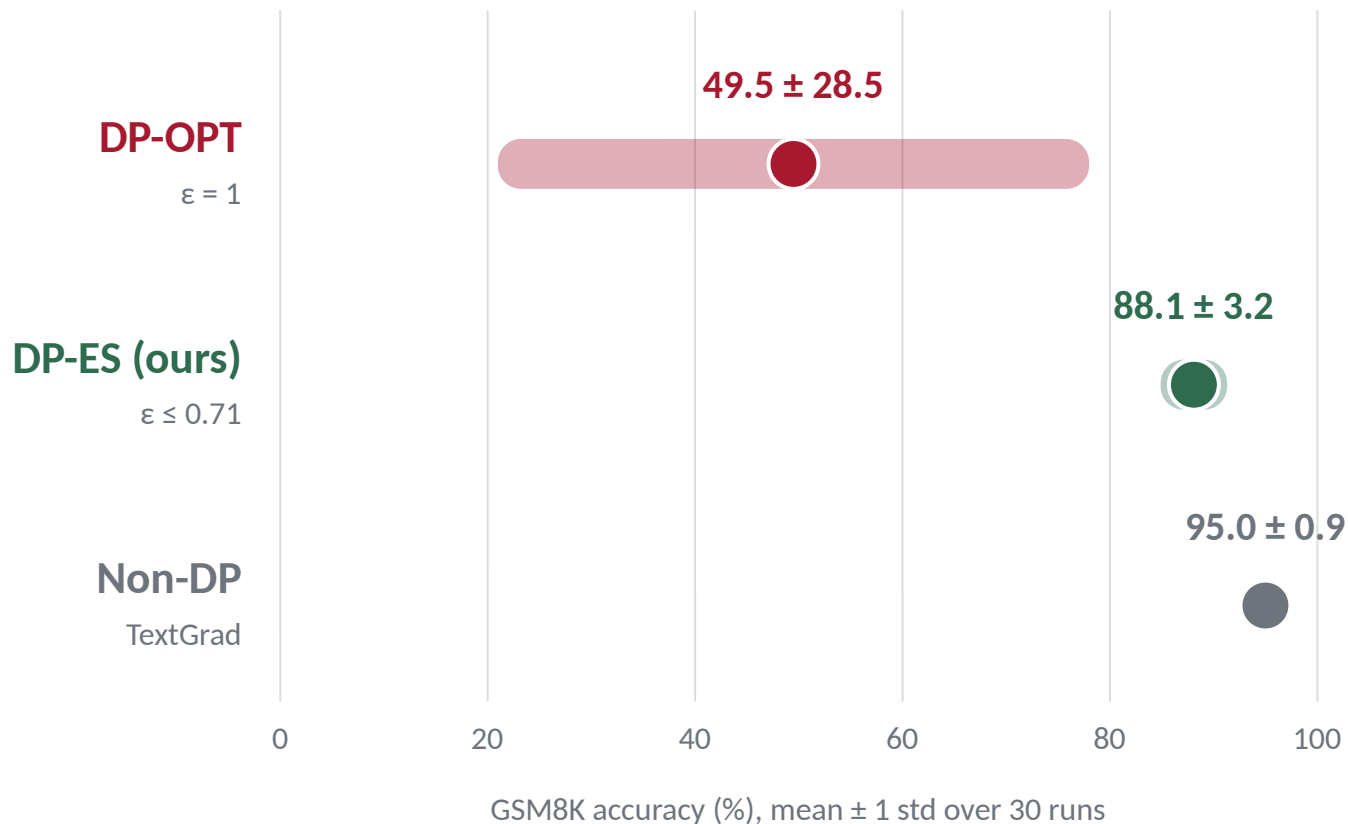
$$\Pr[M(D) \in S] \leq e^\epsilon \cdot \Pr[M(D') \in S] + \delta$$

Privacy unit: one record (x_i, y_i) . Released object: the final prompt. The LLM is a black box reached through an API.

Method	Formal DP	Discrete	Mechanism
TextGrad	✗	✓	LLM “gradients”
OPRO	✗	✓	LLM scoring
EvoPrompt	✗	✓	Evolutionary search
PromptBreeder	✗	✓	Evolutionary search
DP-SGD	✓	✗	Gaussian noise
DP-OPT	✓	✓	Histogram + exp. mech.
DP-ES (ours)	✓	✓	Evolution + Gaussian

DP-ES adds formal record-level DP to evolutionary optimization over discrete prompts.

DP-OPT is unstable on multi-step reasoning



± 28.5 pp

run-to-run standard deviation of DP-OPT on GSM8K: one run may succeed, the next may collapse.

$\sim 9\times$

lower standard deviation with DP-ES (3.2 pp), at a tighter privacy bound.

DP-OPT: black-box adaptation of Hong et al. (ICLR 2024); LLM: DeepSeek-V3.2.

A logged DP-OPT run drifts from its template

Iteration 0 · selected

{question} kept

Carefully solve the following math question and provide a clear explanation of your reasoning: {question}



Iteration 1 · selected

{question} lost

Solve the math problem step-by-step, ensuring each step is clearly explained and justified before proceeding to the next.



Iteration 2 · selected (final)

{question} lost

Present a structured solution by enumerating each step, explaining the purpose of that step, and confirming its correctness before continuing.

21 / 33

candidate prompts (64%) omit the {question} placeholder during search

2 / 3

selected prompts omit it in this run



Caveat. This seed still scores 91%: the harness appends the question when the placeholder is missing. We read omission as structural drift, not as a failure label.

Three structural sources of instability



Token-level myopia

Prompts are built greedily, one token at a time, by voting over private histograms. Nothing checks global coherence, so placeholders can be dropped and text drifts toward generic instructions.



No backtracking

Once a token is released it cannot be revised. An early poor choice is locked in for the rest of construction.



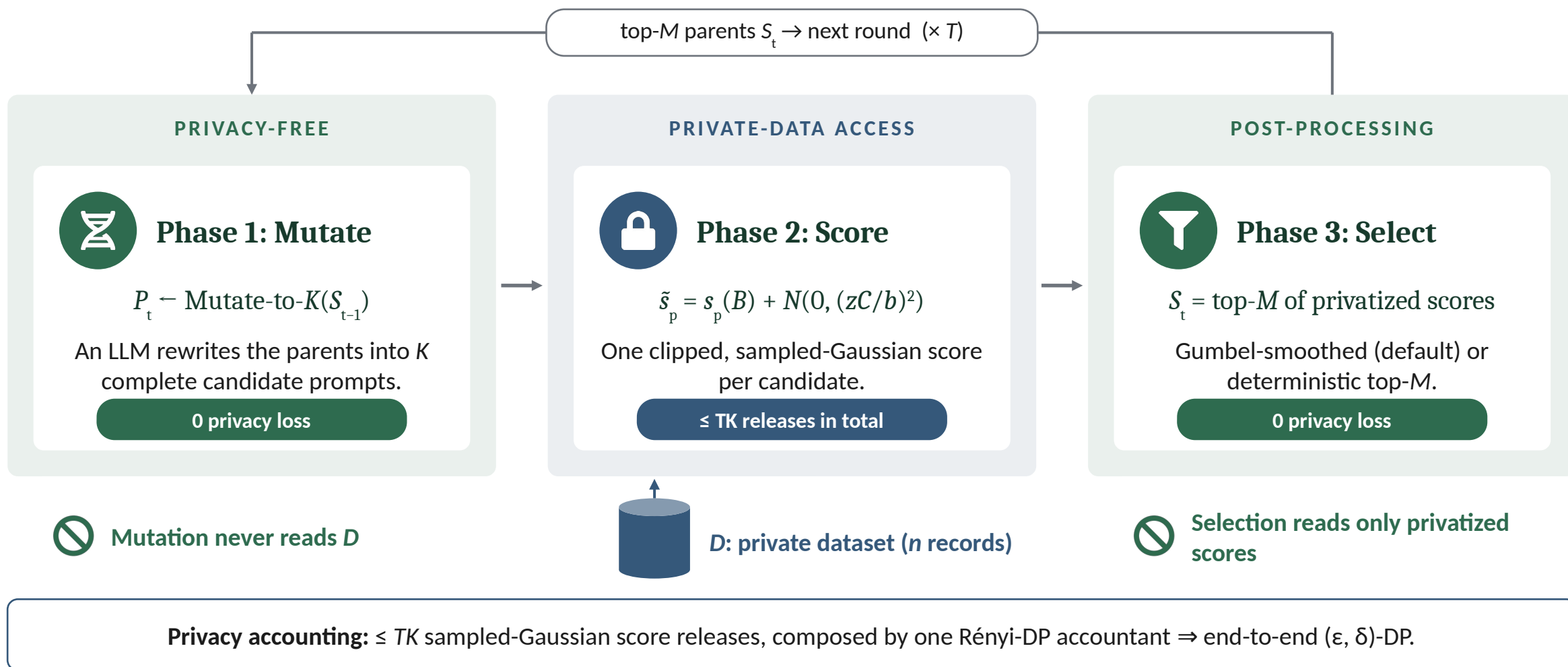
Noise-sensitive selection

DP noise can reverse the ranking of candidates with similar utility, and small evaluation samples (10–20 examples) add variance. With no recovery path, one inversion can redirect the search.



Implication: change the search unit. Evolve *complete* prompts in a population that can recover, and spend privacy only on scoring — the Private Evolution principle (Lin et al., 2024; Xie et al., 2024).

DP-ES: only scoring touches private data



One round: mutate, score privately, select

1 Privacy-free mutation

$$P_t \leftarrow \text{Mutate-to-}K(S_{t-1}), \quad |P_t| = K$$

Sees only parent prompts — never records or record-derived feedback.

2 Sampled-Gaussian scoring

$$u_i(p) = \mathbf{1}[\text{LLM}(p, x_i) = y_i], \quad s_p(B) = (1/b) \sum_{i \in B} \text{clip}(u_i(p), [0, C])$$

$$\tilde{s}_p = s_p(B) + N(0, \sigma^2), \quad \sigma = zC/b$$

Fresh uniform batch $B \subset D$, $|B| = b$, per candidate; sensitivity C/b .

3 Selection over privatized scores

$$\hat{s}_p = \tilde{s}_p + \text{Gumbel}(0, \beta) \quad (\text{default}) \quad \text{or} \quad \hat{s}_p = \tilde{s}_p \quad (\text{deterministic})$$

$S_t \leftarrow \text{top-}M(P_t, \hat{s})$. Both rules are post-processing.

CONFIGURATION (ALL TASKS)

$K = 6$ candidates per round

$M = 2$ parents kept

$T = 3$ rounds

$b = 10$ scoring batch size

$z = 10$ noise multiplier

$C = 1$ clipping bound

$n = 200$ optimization records

Mode BALANCED mutation

Fixed in advance; not selected using task utility.

End-to-end guarantee: $\epsilon \leq 0.71$ at $\delta = 10^{-5}$

Proposition 1**Mutation is privacy-free**

Mutate sees only a public or previously privatized parent prompt, so by post-processing it consumes zero privacy budget.

Proposition 2**Decoupled exploration**

Privacy loss depends only on the score releases — not on the number or strength of mutations per round.

Theorem**End-to-end (ϵ, δ) -DP**

Composing $\leq TK$ sampled-Gaussian releases in Rényi DP (subsampling without replacement) gives (ϵ, δ) -DP for the released prompt.

ACCOUNTING FOR THE REPORTED RUNS

$$n = 200, b = 10 \quad \rightarrow \quad q = b/n = 0.05$$

$$z = 10, C = 1 \quad \rightarrow \quad \sigma = zC/b = 1.0$$

$$K = 6, T = 3 \quad \rightarrow \quad \leq 18 \text{ releases}$$

AutoDP Rényi-DP accountant at $\delta = 10^{-5}$

$$\epsilon = 0.705$$

Reported conservatively as $\epsilon \leq 1$, the setting used for DP-OPT.

Four tasks, one fixed configuration



GSM8K

Multi-step math reasoning

200-example optimization set; full 1,319-example test set



MedQA

Medical multiple-choice QA

200-question USMLE subset; near-ceiling regime



BANKING77

Intent classification

77 classes, 200 examples; large label space



Alpaca

Instruction following

200 instructions; LLM-judge score (DeepSeek-Chat)



Models

- DeepSeek-V3.2 via API (main results)
- Qwen2.5-7B-Instruct, local (validation)



Baselines

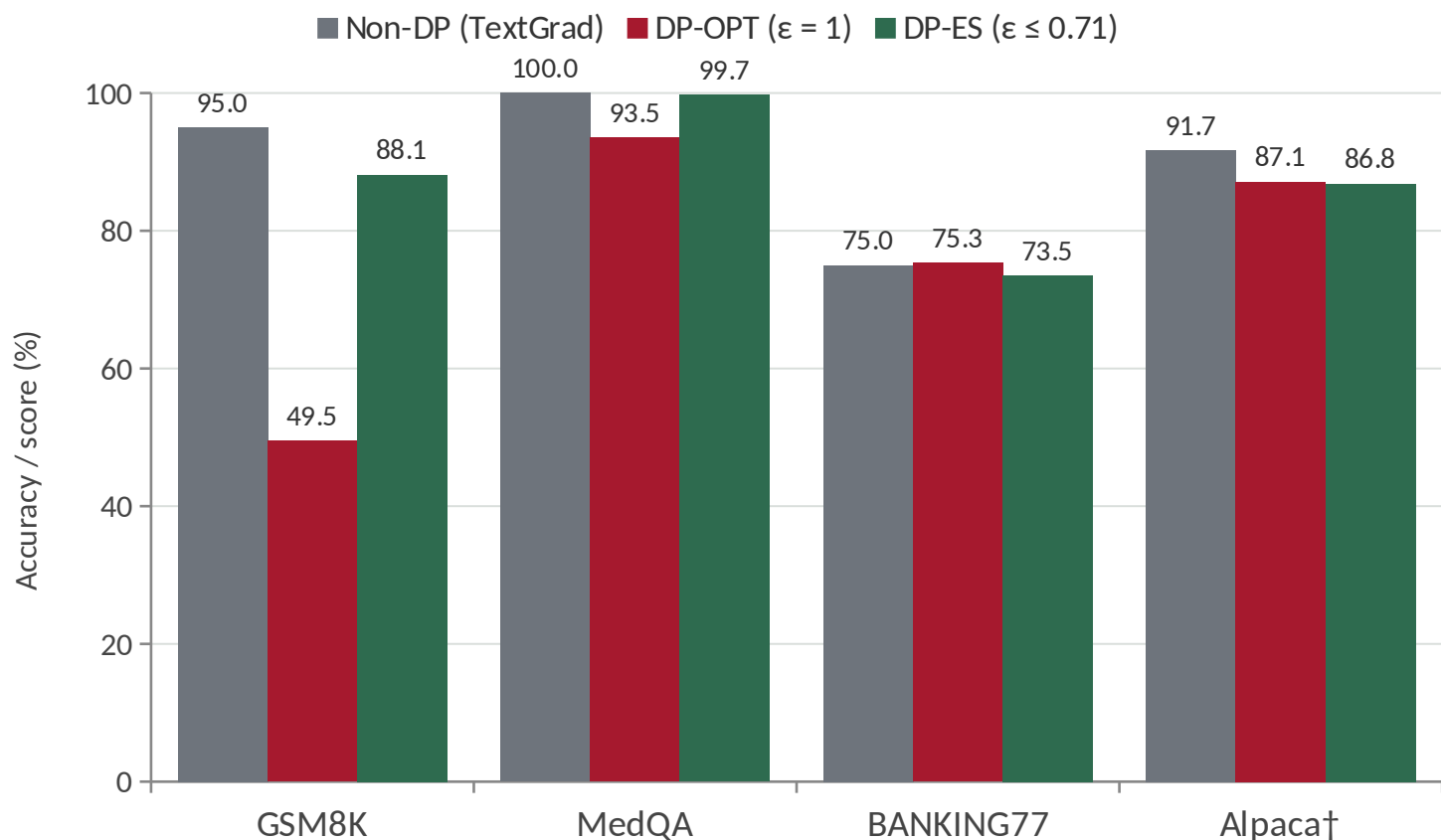
- DP-OPT, black-box adaptation ($\epsilon = 1$)
- Non-DP TextGrad (reference, not a bound)
- PromptDPSGD, soft prompts (local)



Protocol

- Common ceiling $\epsilon \leq 1$, $\delta = 10^{-5}$
- 30 runs on GSM8K; 3 seeds elsewhere
- Same initial prompt for all methods

Largest gains where prompt structure matters



Mean over 30 runs (GSM8K) or 3 seeds; $\delta = 10^{-5}$. †LLM-judge score scaled to %.

DP-ES - DP-OPT (pp)

GSM8K **+38.6**

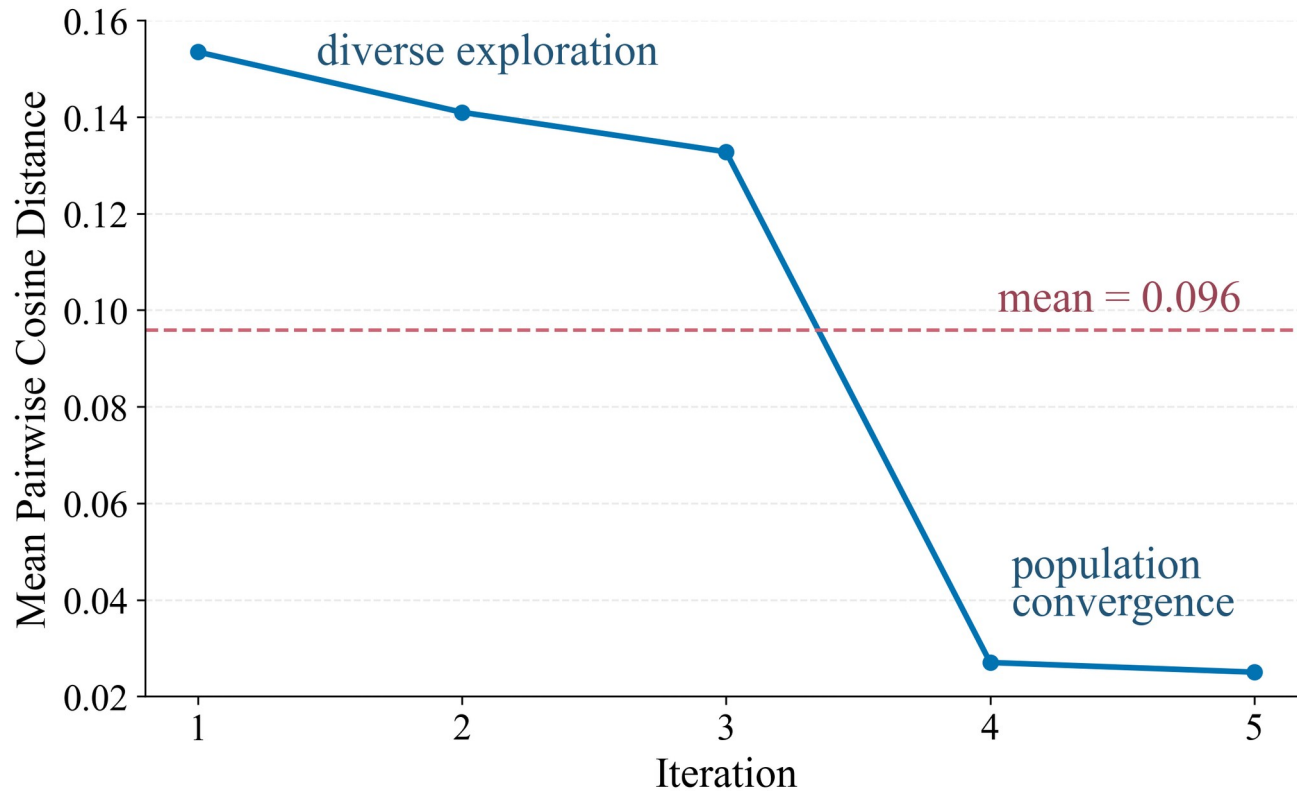
MedQA **+6.2**

BANKING77 **-1.8**

Alpaca **-0.3**

Large gains on reasoning-heavy tasks; parity on well-structured classification and instruction following. MedQA is near ceiling.

Populations retain alternatives under noise



Population diversity: mean pairwise cosine distance of prompt embeddings (all-mpnet-base-v2) across iterations.

Final DP-ES prompt (GSM8K)

{question} kept

You are a math tutor. Solve the question carefully and show reasoning: {question} (Provide detailed reasoning and justification.) (Be concise and accurate.)



Mutations edit *complete* strings, preserving global prompt structure.



Six candidates per round supply alternatives after noise-induced rank inversions.



Exploration is free: richer mutation costs no privacy budget.

2.5× faster, 3.3× fewer private-data calls

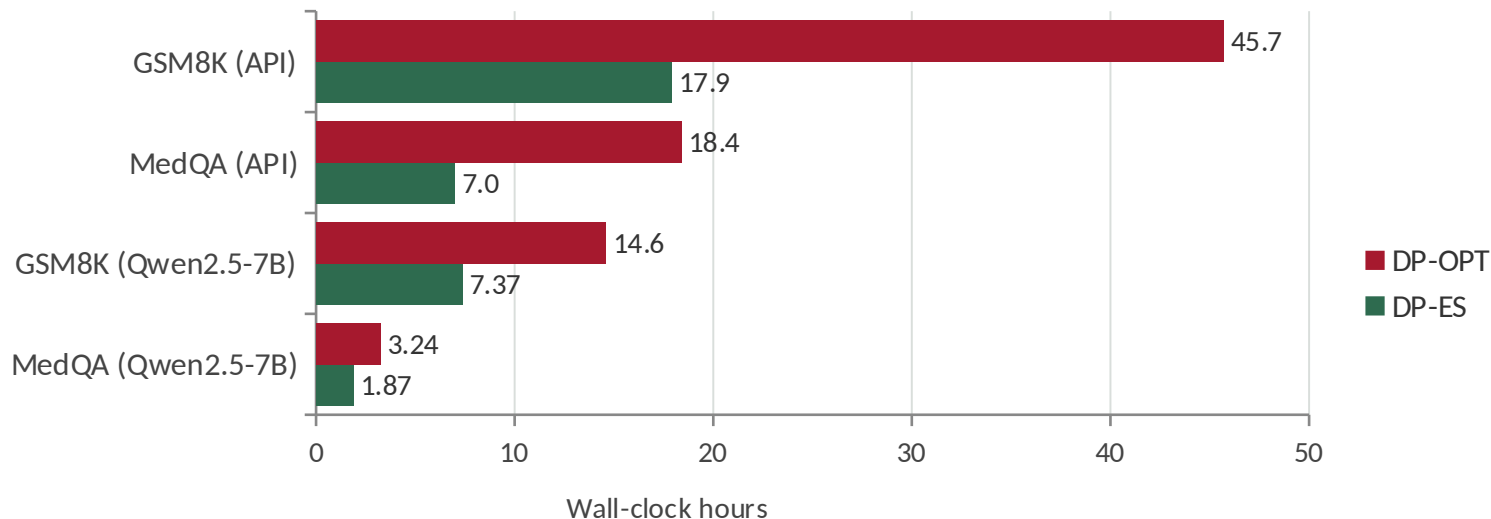
2.5×

faster wall-clock

17.9 h vs. 45.7 h on GSM8K (API)

3.3×fewer private-data
call groups

10 vs. 33 in a representative run

2.8×more tokens —
the trade-off11,120 vs. 4,016: DP-ES trades tokens
for robustness

Why faster?

DP-OPT queries private aggregates both while constructing tokens and while evaluating. DP-ES mutations never touch D , so far fewer serial private-data round trips are needed — latency drops even though token count rises.

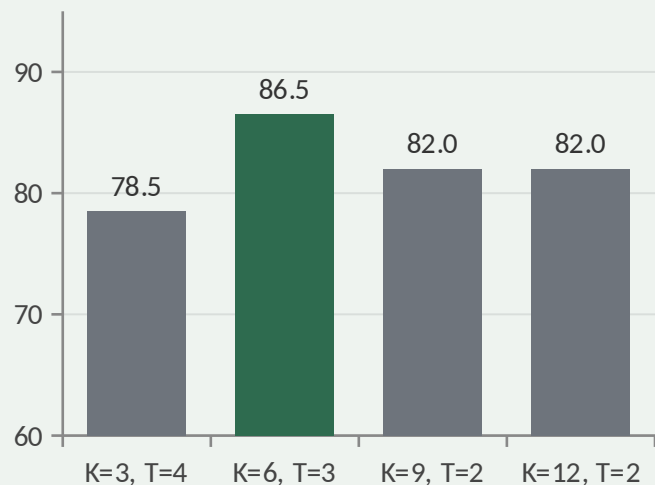
Robust to selector, population size, and model

(a) Selection rule

Selector	GSM8K (%)
Gumbel-smoothed	88.8 ± 3.4
Deterministic	88.6 ± 2.9

Five seeds. Both rules post-process privatized scores, so the privacy guarantee is identical; accuracy differs by only 0.2 pp.

(b) Population size ($K \cdot T \leq 24$)



$K = 6, T = 3$ balances diversity and depth: small K explores too narrowly, large K leaves too few refinement rounds.

(c) Local Qwen2.5-7B ($\epsilon \leq 1$)

Method	GSM8K	MedQA
DP-ES	53.8 ± 6.1	81.2 ± 1.3
DP-OPT	37.4 ± 11.6	83.0 ± 2.1
PromptDPSGD	20.0 ± 5.8	34.8 ± 19.0
Non-DP	26.2 ± 10.2	79.5 ± 4.2

Three seeds. The reasoning gain transfers (+16.4 pp on GSM8K); MedQA is comparable, matching the API trend.

Stress test: no listed PII strings in DP-ES prompts



Setup. 200 synthetic customer profiles × 6 sensitive strings (name, phone, email, full address, street fragment, account ID) = **1,200 strings**. We count exact and near-verbatim matches in the final prompt.

Adversarial non-DP control

Given raw records and instructed to embed a directory

```
Use this customer directory:  
Jessica Moore, (713) 907-8800,  
jessica.moore83@mail.com;  
Anthony Williams, (312) 991-9095,  
anthony.williams60@mail.com; ...
```

40

**strings leaked
(3.33%)**

DP-ES

Mutation never receives records

```
You are a customer service  
assistant. Retrieve relevant  
information based on user queries.  
If the query involves account  
information, verify before  
responding.
```

0

**strings leaked
(0.0%)**



Also: 1,000 logged noise draws are consistent with the configured Gaussian (Kolmogorov–Smirnov). These are diagnostics — they cannot certify DP; the guarantee comes from the RDP accountant.

What we establish — and what remains open

Supported by the evidence

- ✓ More robust optimization under DP noise than token-level construction
- ✓ A formal record-level guarantee ($\epsilon \leq 0.71$, $\delta = 10^{-5}$) from mechanism-level RDP accounting
- ✓ Lower wall-clock time and fewer private-data round trips than DP-OPT

Open — future work

- ? End-to-end validation on genuinely sensitive, non-saturated deployment data
- ? MedQA is near ceiling (Non-DP 100.0%), limiting its discriminative power
- ? Attacks: exact-match only — no membership inference or reconstruction
- ? One privacy level ($\epsilon \leq 1$) rather than a dense privacy-utility frontier
- ? Advantage is task-dependent: parity on BANKING77 and Alpaca

Next steps: adaptive population sizing · multi-objective search (utility, privacy, stability) · federated DP-ES for distributed sensitive data

Takeaways

01

Diagnosis

Token-level DP prompt construction is structurally fragile: $49.5 \pm 28.5\%$ on GSM8K over 30 runs, with template drift.

02

Method

DP-ES evolves complete prompts; only sampled-Gaussian scoring spends privacy, while mutation and selection are post-processing.

03

Evidence

88.1% on GSM8K (+38.6 pp, $\sim 9\times$ lower std) and $2.5\times$ faster, under $\epsilon \leq 0.71$, $\delta = 10^{-5}$.

Thank you — questions are welcome.



Code, accountant and paper source: github.com/StephCpa/dp-es

Contact: liaiping@nudt.edu.cn

Threat model and assumptions



Trusted curator

The party running DP-ES may access D but releases only the final prompt.



Record-level privacy

The unit is one record (x_i, y_i) ; group privacy scales ϵ with group size.



Trusted boundary

The evaluation endpoint is a trusted processor; DP protects the released prompt, not API inputs.



Mutation isolation

The mutation LLM sees only parent prompts. PII memorized in pretraining is outside this threat model.



Post-processing safety

Downstream use of the released prompt adds no further loss about D .



Audits are not proofs

Empirical checks can expose implementation errors but cannot certify DP; the guarantee is mechanism-level.

Where the efficiency comes from

Cost component	DP-OPT	DP-ES	Ratio
<i>Logged private-data call groups</i>			
Mutation (token-level / privacy-free)	3 (DP)	0 (free)	—
Evaluation (Gaussian-noised)	30	10	3.0× fewer
Total private-data groups	33	10	3.3× fewer
<i>Token usage</i>			
Input tokens	1,309	6,120	4.7× more
Output tokens	2,707	5,000	1.8× more
Total tokens	4,016	11,120	2.8× more
<i>Wall-clock</i>			
Runtime (hours)	16.6	6.5	2.5× faster
Accuracy (single run)	91.0%	86.5%	—

GSM8K, seed 42, $\epsilon \leq 1$, 200-example optimization set. Counts are logger-level call groups, not individual requests.

Reading the table

DP-OPT's token construction queries private counts, so its mutation calls also enter the accountant. DP-ES mutation is post-processing; only its evaluation groups touch records.

Fewer serial round trips dominate latency, so runtime drops even though DP-ES uses more tokens.

Reproducing the privacy bound

```
$ python -m pip install -e .  
$ python scripts/compute_privacy.py  
epsilon=0.705168  
delta=1e-05  
releases=18  
sampling_rate=0.050000
```

Defaults: $n = 200$, $b = 10$, $z = 10$, $K = 6$, $T = 3$, $\delta = 10^{-5}$. Recompute whenever any of them changes: larger KT raises both cost and composed privacy loss.

Privacy contract of the released code

- 1 Per-record utilities are clipped to a public interval.
- 2 Every data-dependent evaluation uses the sampled-Gaussian scorer; raw utilities are never released.
- 3 The number of releases is fixed in advance.
- 4 Mutation receives no private records or record-derived feedback.
- 5 Selection and all later computation use only privatized outputs.